Pr. 4 - calculative Foundation

1) Represent each student's subject scores as a vector.

→ if a student has scores in n subjects, the scores can be represent as a vector: $V = [80, 75, 90]$.

Each element represent the score ~~of~~ of one subject.

(2) Compute Norm-1 and Norm-2 of vectors.

→ Norm-1: Sum of the absolute values

$||V||_1 [x_1] + [x_2] + \dots + [x_n]$

Norm-2: square root of the sum ~~of~~ of squared values.

$||V||_2 = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$

• Dot product and angle Between two Student's score vectors

→ ~~Dot pro~~ The dot product of two vectors is.

$A \cdot B = \Sigma A_i B_i$

The angle Between them is:

$\theta = \cos^{-1}[(A \cdot B)/(||A|| \, ||B||)]$.
it shows how similar the two vectors are.

- cross product for 3D selected subjects.

→ The cross product is used for two 3D vectors.
$A \times B$.
it produces a new vector that is perpendicular to Both A and B.

- projection of one vector onto another.

⇒ The projection of vector ~~A~~ onto vector B is:
$proj^{B}(A) = (A \cdot B / B \cdot B) B$

it gives the ~~part~~ part of A that ~~line~~ lies in the direction of B.

Part B: Matrix operations.

(1) Form a matrix of students x subjects perform matrix addition and multiplication.

- Students scores can be represented using a matrix ~~where~~ where:
- Rows = students.
- columns = subjects.

Example:

A = [[80, 75, 90],
[70, 85, 88]].

Matrix Addition: Add corresponding elements.
Matrix Multiplication: Multiply rows of the first matrix with columns of the second matrix.

- Transpose and inverse.

→ it is represented as AT.
inverse: A matrix inverse is represented as A-1.
it exists only when the matrix is square and:
$det(A) \neq 0$
Also
$A \times A^{-1} = 1$.

• Determinant

→ The Determinant is a single numeric value calculated from a square matrix.
it helps determine whether a matrix has an inverse.

if $det(A) \neq 0 \rightarrow$ inverse exists.
if $det(A) = 0 \rightarrow$ inverse does not ~~exist~~ exist.

C. Linear Transformations & Geometry.

(1) Explain line plane, and hyperplane.

→ • Line: A 1-dimensional object.
• Plane: A 2-dimensional flat surface.
• Hyperplane: A (n-1)-Dimensional surface in n-Dimensional space.

Hyperplanes are commonly used to separate data in machine learning

(2) Dimensionality from 2D-3D-higher dimensions

→ 2D: Two features - a line can separate data.

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• 3D: Three features → a plane can separate data.

• Higher Dimensions: More features → a hyperplane can separate data.

Thus, dimensionality ~~increases~~ increases as the ~~number~~ number of features ~~incr~~ increases.

Part D: Eigenvalues & Decomposition

(1) Compute eigenvalues and eigenvectors of the covariance matrix:

→ For a covariance matrix C, eigenvalues and ~~eigen~~ ~~eigenvalue~~ eigenvectors satisfy:

$$Cv = \lambda v$$

Where:
- λ = Eigenvalue
- V = Eigenvector

Eigenvectors represent the main directions of variation in the data.

Eigenvalues represent the amount of variance in those directions